Impact energy and TNT equivalents

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Convert an energy estimate without claiming a complete damage prediction.

SUCCESS CRITERIA - I can...

1. Convert speed units before squaring.
2. Calculate kinetic energy from the supplied mass and velocity.
3. Distinguish an energy equivalent from an explosion or damage forecast.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “equivalent” and “kinetic energy” to explain your answer. Sentence starters are optional.

equivalent: equal in the specified quantity, not necessarily in other effects

kinetic energy: energy associated with motion

impact: a collision or strike between objects

01 Warm-up: bring back an earlier idea

Two students perform the same lift. One takes **3** times as long as the other. **Compare work** and average power.

Word help: power = rate of energy transfer

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $K = \frac{1}{2}mv^2$; **1 ton** TNT equivalent = **4.184×10^9 J**. Equivalent energy does not mean identical effects.

Worked reasoning: A fictional event transfers **8.4×10^6 J**. Using **4.2×10^6 J** per kg TNT as an energy comparison gives **2 kg** TNT equivalent. This compares energy, not identical effects.

Your turn: Draw a labeled energy comparison and **name one effect this number** alone cannot predict.

Sentence starter: I predict ... . I used ... to make this prediction because ... .

Initial **Transfers** **Final**

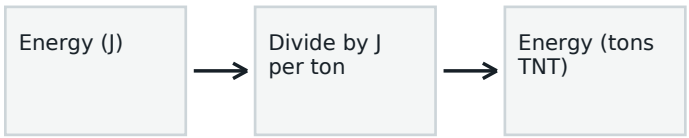
Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Two reports describe the same fictional impact: one uses joules and the other uses TNT-equivalent tons.
1. Use the conversion diagram to **explain why the units can** express the same quantity.
2. What does this comparison leave out?

Sentence starter: I think ... . I can point to ..., which shows ..., so ... .



Marzano check - circle: ① ② ③

04 Apply the physics

A supplied energy is **8.368×10^9 J**. Divide by **4.184×10^9 J**/ton.
1. **Find TNT equivalent**.
2. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Sentence starter: Both ... . They differ in ... because ... .

Energy (J)	J per ton TNT	Equivalent tons
___	4.184×10^9	___

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Why should an energy-equivalent number not be reported as a precise area of destruction?

Sentence starter: I think … . I can point to …, which shows …, so … .

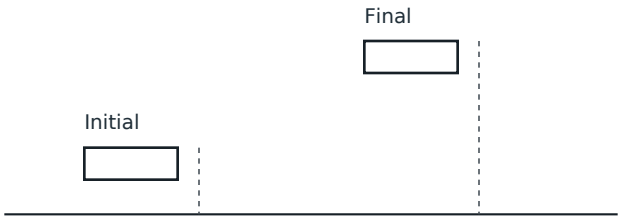
Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Lift a **5 kg** object by **7 m**. Use g=**10 N/kg**. **Find the change** in gravitational potential energy.
Label the distances or energy levels used in your reasoning.

Word help: potential energy = energy associated with the configuration of interacting objects

Sentence starter: I found … [unit]. This value means … in this situation.



Choose and label a zero-height level.

Marzano check - circle: 1 2 3

07 Exit check

At fixed mass, speed halves. **What happens** to K and its TNT equivalent? **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Sentence starter: Both … . They differ in … because … .



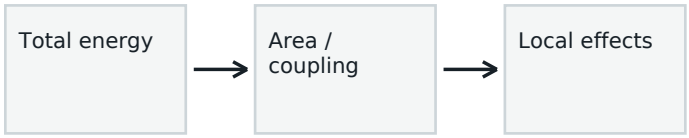
Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Two events have equal energy but spread it over different areas.
1. **Explain why local effects could** differ without changing total energy.
2. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: 1 2 3